AWS CloudWatch Basic

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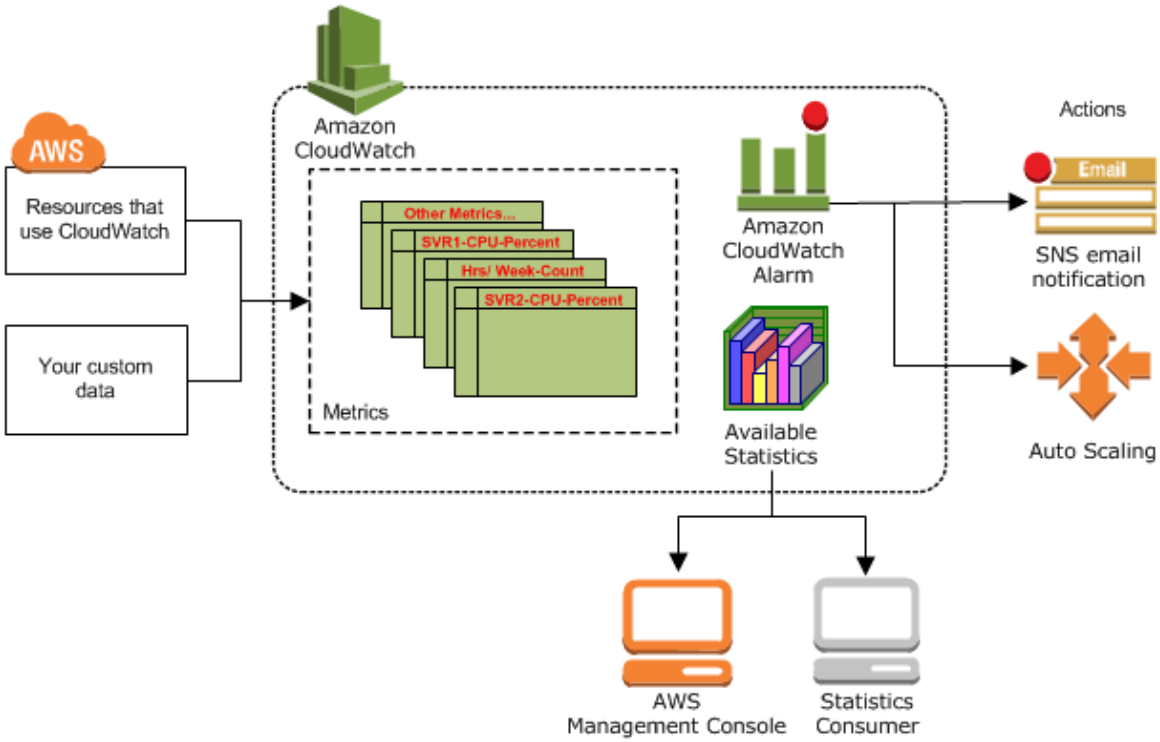
Inntroduction of CloudWatch:

Amazon CloudWatch monitors your Amazon Web Services (AWS) resources and the applications you run on AWS in real time, and offers many tools to give you system-wide observability of your application performance, operational health, and resource utilization.

Simple Monitoring Architechture



Benefits of CloudWatch:

Centralized Monitoring: View all AWS resources, apps, and on-premises servers in one place.

Real-Time Metrics & Logs: Get near real-time visibility into performance, errors, and usage.

Automated Alerts & Scaling: Use alarms to send notifications or trigger Auto Scaling actions.

Improves Reliability: Detect failures quickly and troubleshoot using logs, metrics, and dashboards.

Cost Optimization: Analyze usage patterns to optimize resource allocation and reduce costs.

CloudWatch Features:

Metrics: Collects performance metrics from AWS services (e.g., CPU utilization, network traffic) and provides a central repository for them.

- **Custom Metrics:** You can publish your own application-specific metrics to CloudWatch.
- **Metric Streams:** Allows for continuous, near real-time streaming of metrics to a destination of your choice.

Logs: Enables you to centralize, monitor, and analyze log files from various AWS resources, applications, and on-premises servers.

- **Log Groups and Streams:** Logs are organized into groups and streams for easier management and analysis.
- **Metric Filters:** Extracts numerical data from logs and transforms it into CloudWatch metrics.

Alarms: Watches a single metric over a specified time period and takes one or more actions based on the value of the metric.

- **Actions:** Can trigger notifications (via Amazon SNS), and automated actions like scaling resources with Auto Scaling or stopping an EC2 instance.
- **Composite Alarms:** Combines multiple alarms to reduce noise and provide a more holistic view.

CloudWatch Features:

Dashboards: Provides customizable visualizations of your metrics and alarms in a single view.

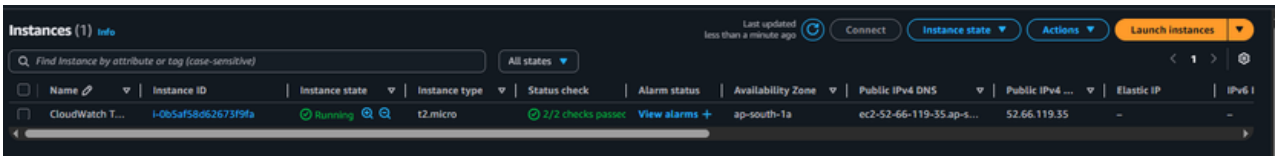
- **Widgets:** You can add various widgets (e.g., line graphs, numbers, gauges) to display key performance indicators (KPIs) and monitor the health of your environment at a glance.
- **Cross-Account Observability:** Allows you to view metrics and logs from multiple AWS accounts on a single dashboard.

Events and Insights:

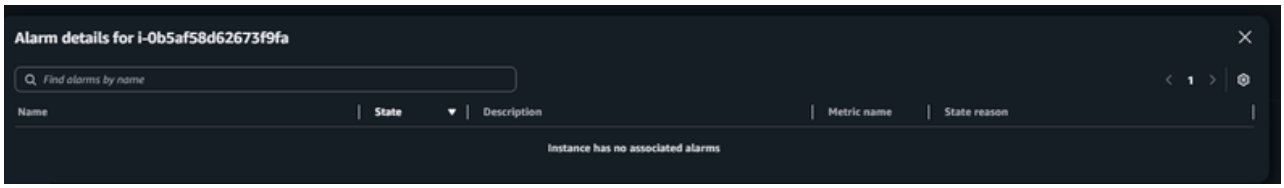
- **Events (Amazon EventBridge):** Delivers a near real-time stream of system events that describe changes in your AWS resources. You can create rules to automatically route these events to a target.
- **Logs Insights:** A powerful query service for interactively analyzing log data in near real-time, helping you troubleshoot operational issues and identify trends.
- **Container Insights:** Collects, aggregates, and summarizes metrics and logs from containerized applications and microservices (e.g., on Amazon EKS, Amazon ECS, and Kubernetes).
- **Contributor Insights:** Analyzes log data to find the top "contributors" to a metric, helping you identify what is causing a problem.

Hands-on EC2 instance monitoring using CloudWatch Alarms:

To attach CloudWatch Alarms, you first need to create an EC2 instance.

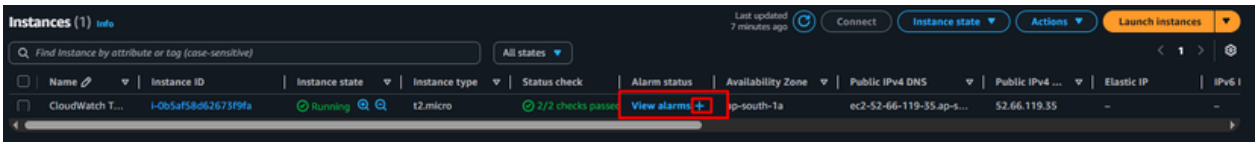


By default, an instance doesn't have any alarms.



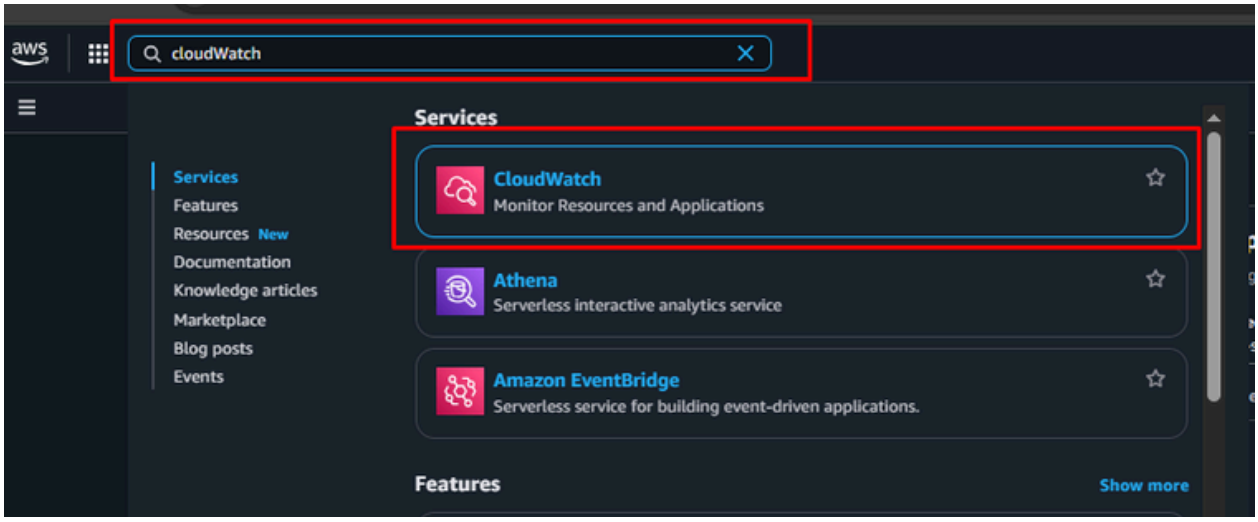
Now it's time to create an Alarm. You can create it from the instance or from the Alarms section in CloudWatch. In my case, I will create it from CloudWatch.

Create directly from instance:

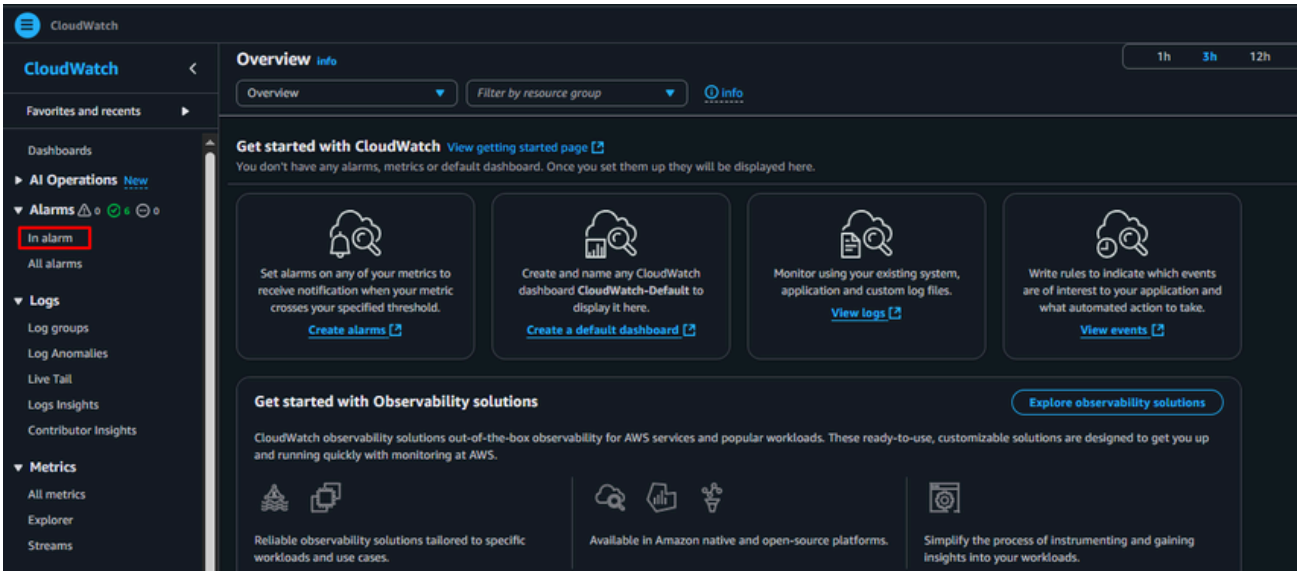


Hands-on EC2 instance monitoring using CloudWatch Alarms:

Create from CloudWatch:
First, search for CloudWatch in your console and click on it.

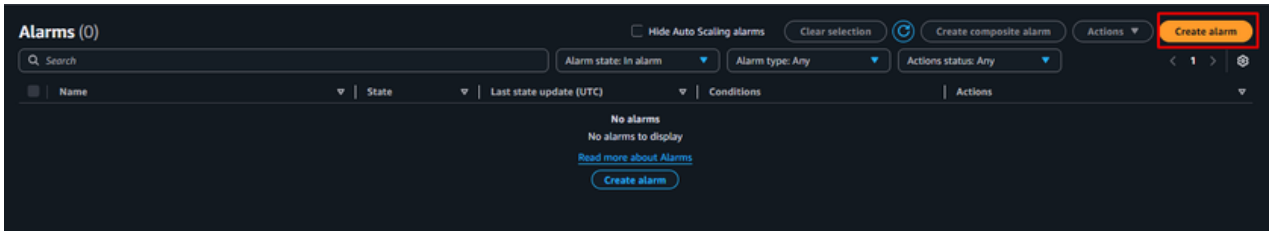


Now click In alarm section.

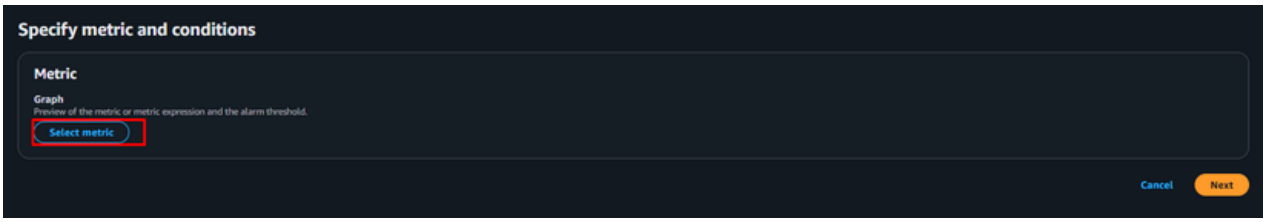


Hands-on EC2 instance monitoring using CloudWatch Alarm:

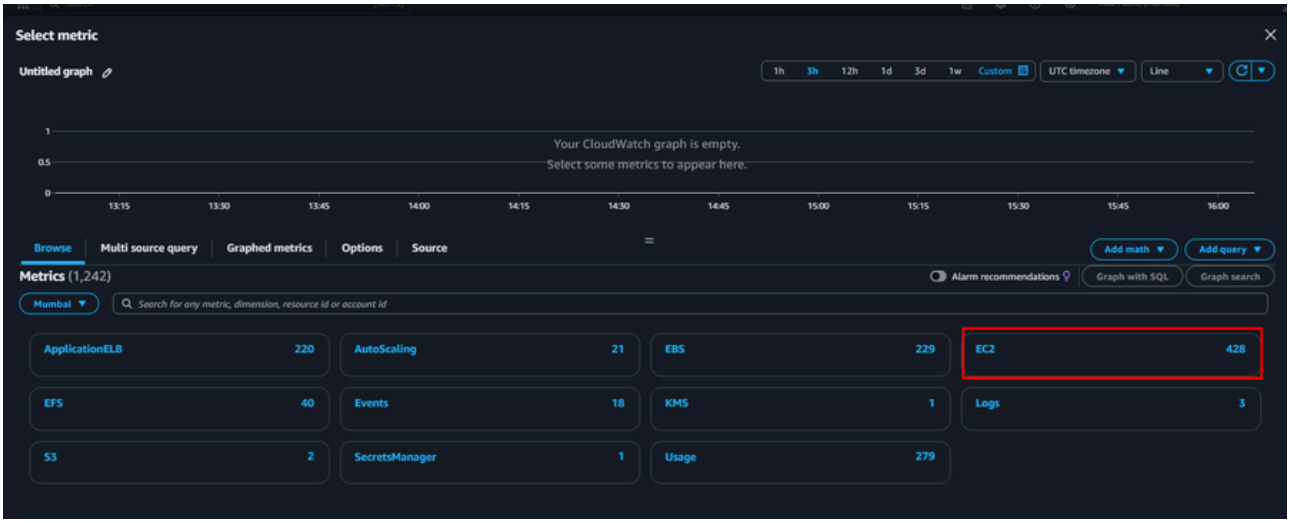
Get this page, now click on create alarms.



Step 1: Select a metric, click on it.

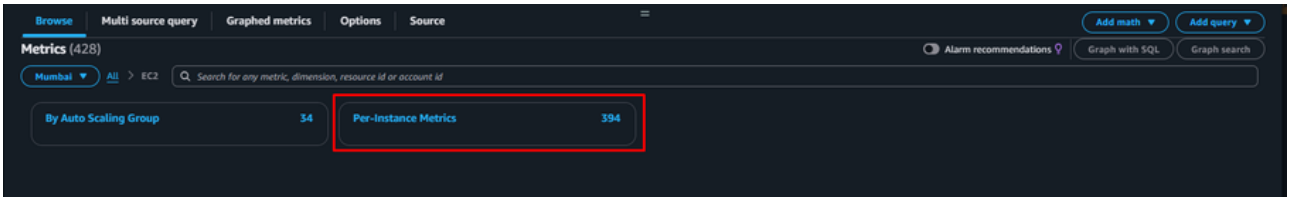


Step 2: I'm creating alarms for EC2 that's why I select EC2.

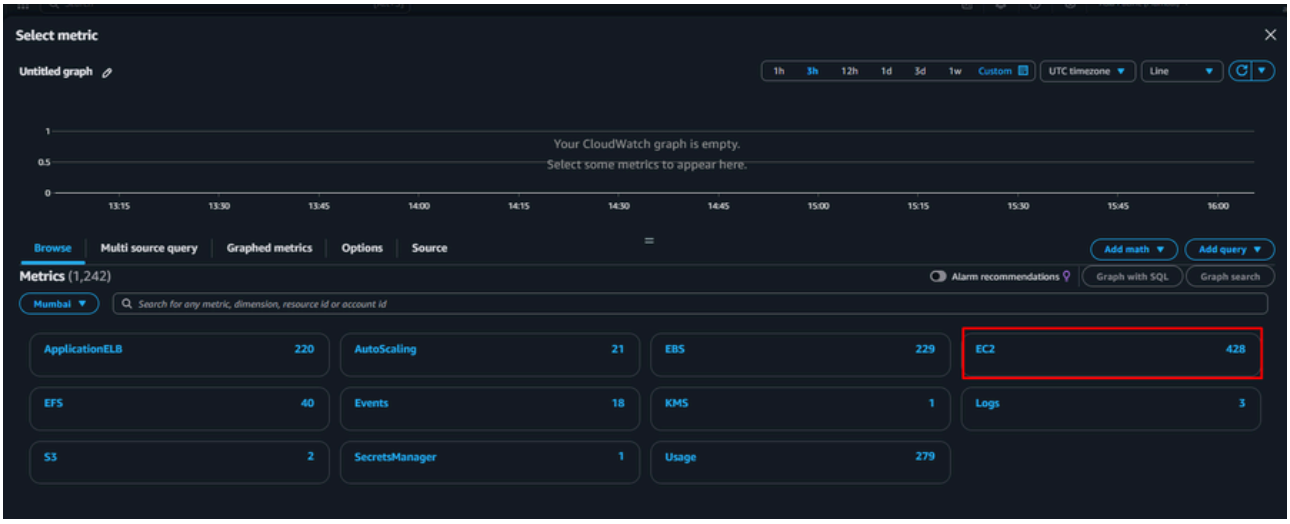


Hands-on EC2 instance monitoring using CloudWatch Alarm:

Step 3: Choose based on your needs. In my case, I selected Per-instance metrics, now click on it.

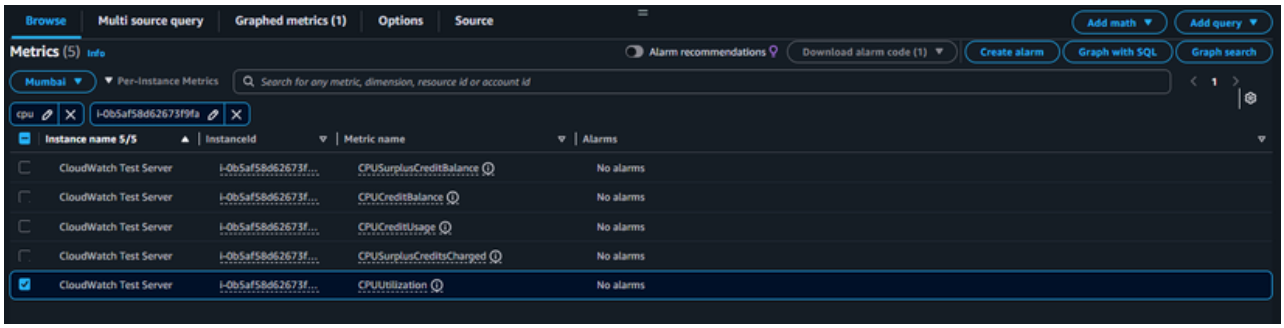


Step 4: Shows this section now choose the service for your metrics i choose EC2.

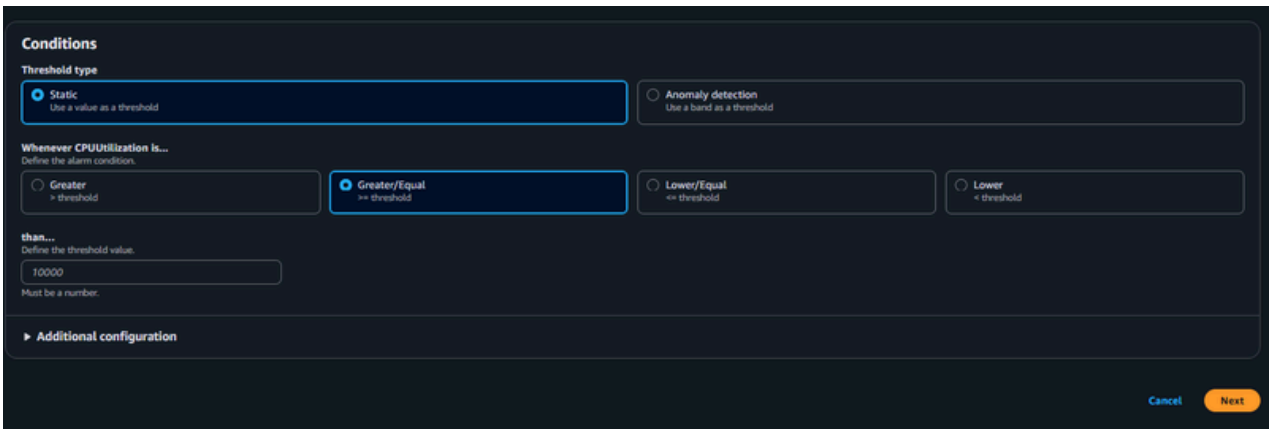


Hands-on EC2 instance monitoring using CloudWatch Alarm:

Step 5: Now, search for your EC2 instance by its instance ID and select the metric you want to add. I'm creating this metric for CPU Utilization monitoring, so I selected that.

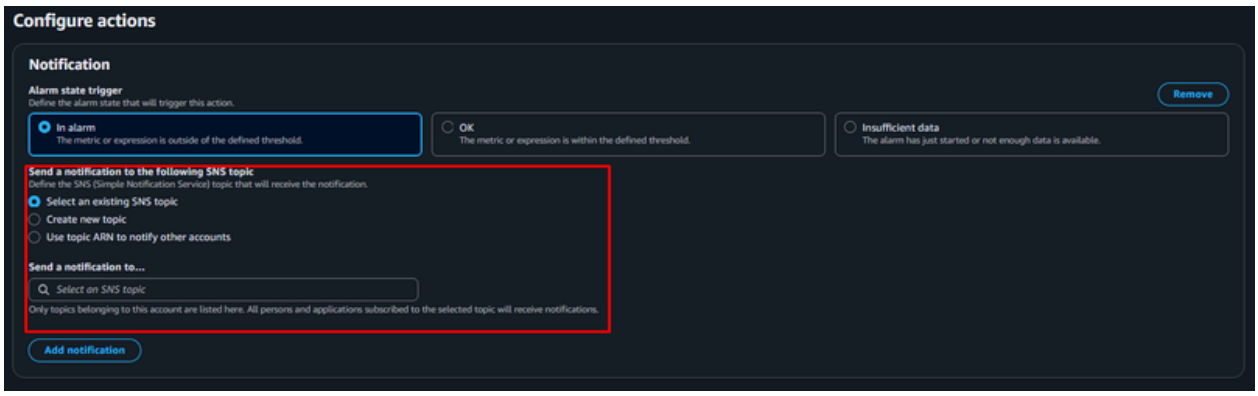


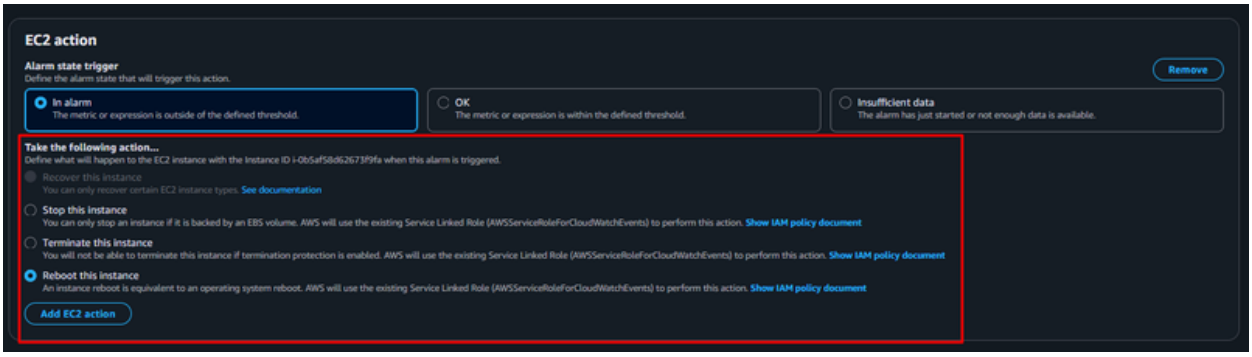
Step 6: In condition section, select Threshold type and Whenever CPUUtilization condition. Don't forget to set threshold value is required.



Hands-on EC2 instance monitoring using CloudWatch Alarm:

Step 7: Choose or create a SNS for getting notification. And Select action.

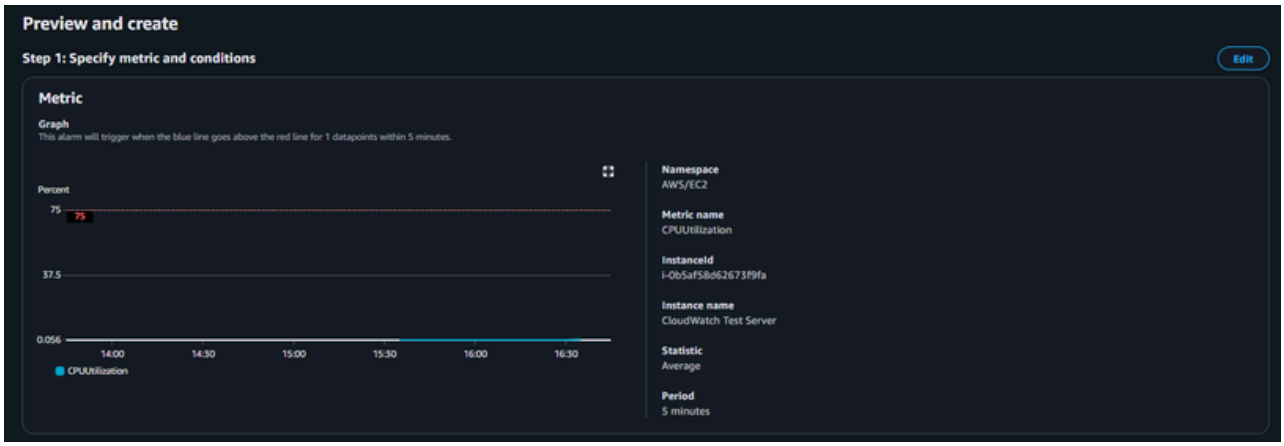




Step 8: In this step, type a name for your alarms.

Hands-on EC2 instance monitoring using CloudWatch Alarm:

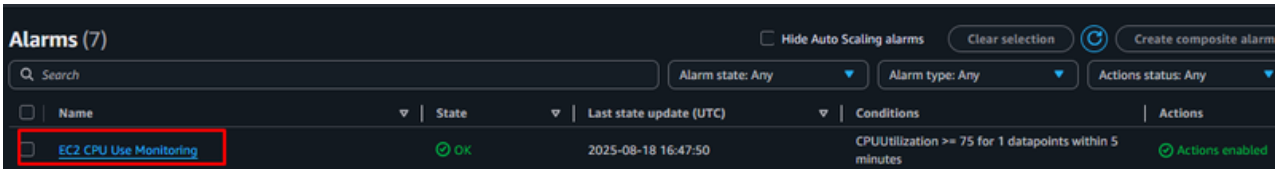
Step 9: Now, review your configuration and create it.



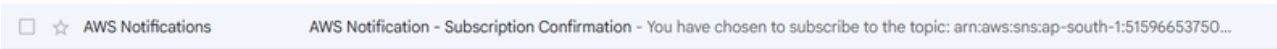
Step 8: In this step, type a name for your alarms.

Hands-on EC2 instance monitoring using CloudWatch Alarm:

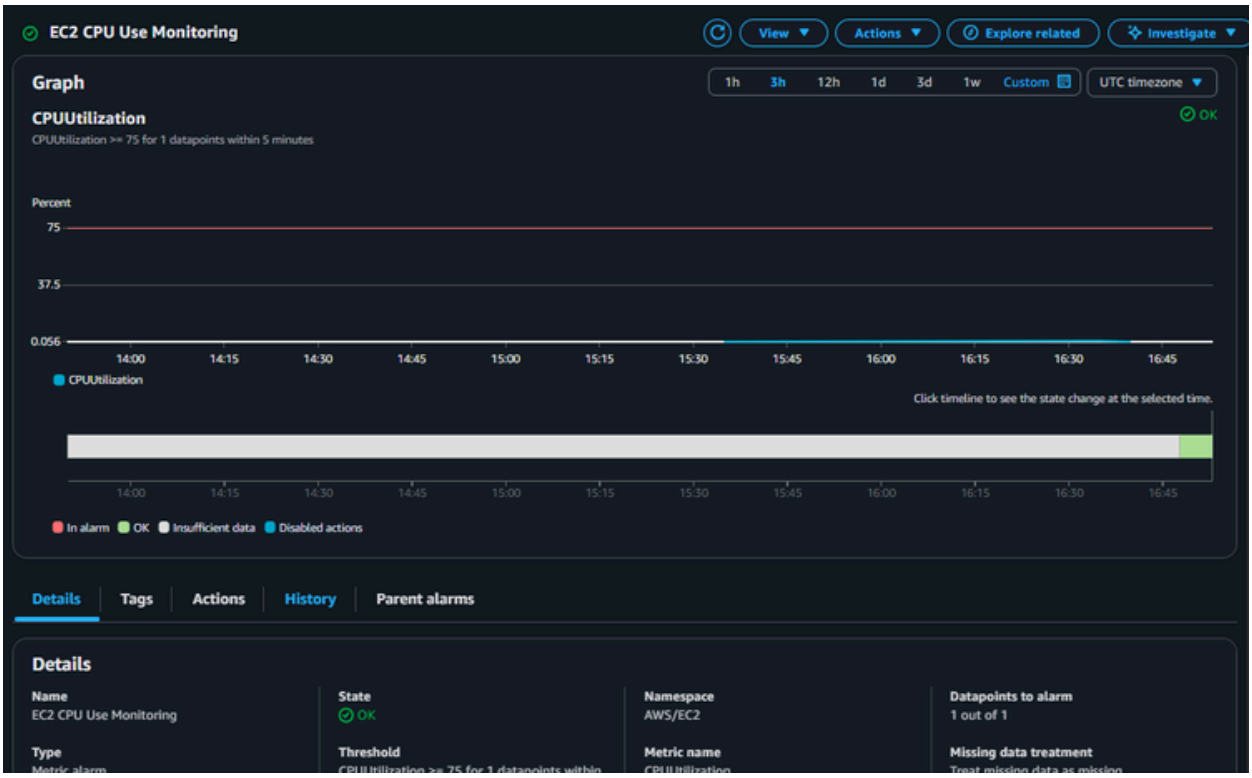
Step 10: Our alarm is created successfully.



You also get a email which you use for SNS.

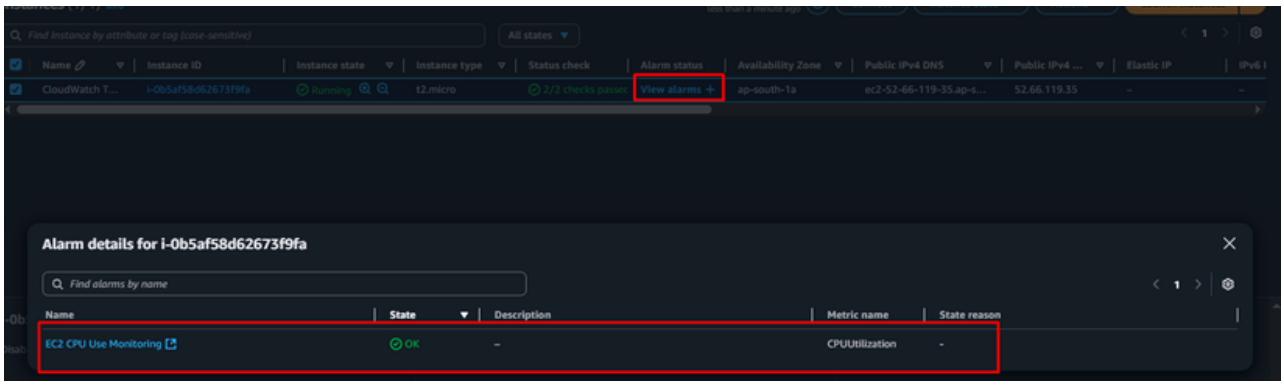


Now, you can view the details of your alarm.



Hands-on EC2 instance monitoring using CloudWatch Alarm:

Go back to your EC2 instance and verify that it is attached successfully.



In my case, it shows the newly created Alarm, which means we successfully created it. Now it is monitoring CPU utilization based on the Alarm configuration. If any issue occurs, it will send an email and reboot the instance, as I set earlier.

I set it to 75% CPU utilization — when this threshold is reached, it will reboot the instance and send me an email via SNS(Simple Notification Service).

Thank You

Stay Connect:

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/alamgirweb11